University of Colombo School of Computing

SCS 1308 – Foundations of Algorithms

Take – Home 01

Instructions

- Try the following questions and upload your answer script as a zip file to the given link in the UGVLE on/before 18th October at 10 PM.
- It's an individual task. The deadline is 10:00 PM. Submissions will be accepted until midnight with a late penalty of 5% deducted for each hour past the deadline.
- Note: Rename your zip file as indexNo_1_last_namefname.zip.

Part - A

1. Consider the following problems:

    a. Finding a student ID in a list containing 2,000 student records.
    b. Processing a matrix containing 50 rows and 80 columns.
    c. Processing a graph containing 100 vertices and 250 edges.

    i. What would be an appropriate measure of input size for each problem?

    ii. Explain why the input size is not necessarily the numerical value of the input.

2. Two students implement the same algorithm.
    - Student A's program takes **0.8 seconds**.
    - Student B's program takes **1.4 seconds**.

    Student A concludes that their algorithm is more efficient. **Do you agree? Explain why or why not.**

3. Consider a sequential search through the following list:

$$[14, 27, 31, 45, 53, 68, 72, 81]$$

The algorithm compares the search key with elements from left to right. Determine the number of comparisons for:

a. Searching for 14
b. Searching for 68
c. Searching for 100

Identify which represents the best and worst cases.

Part - B

4. Determine the asymptotic complexity of the following code.

```
for (i = 1; i <= n; i++)
{
   x = x + 5;
}
```

5. Determine the time complexity of the following algorithm.

```
for (i = 0; i < n; i++)
{
   for (j = 0; j < 5; j++)
   {
      x = x + 1;
   }
}
```

6. Determine the time complexity.

```
for (i = 0; i < n; i++)
{
   for (j = 0; j < n; j++)
   {
      x++;
   }
}
```

7. Determine the overall time complexity.

```
for (i = 0; i < n; i++)
{
   x++;
}
for (j = 0; j < n; j++)
{
   y++;
}
```

8. Determine the time complexity.

```
i = 1;
while (i < n)
{
   x++;
   i = i * 2;
}
```

Part - C

9. Consider:

```
x = 0;
for (i = 0; i < n; i++)
{
   for (j = 0; j < m; j++)
   {
      x = x + 1;
   }
}
```

a. Identify a suitable barometer operation.

b. How many times is it executed?

c. Give the asymptotic complexity.

10. Consider:

```
for (i = 0; i < n; i++)
{
   for (j = 0; j < i; j++)
   {
      x++;
   }
}
```

Determine the number of executions of x++ and its asymptotic complexity.

11. Consider:

```
if (n == 0)
{
   x = 10;
}
else
{
   for (i = 0; i < n; i++)
   {
      x++;
   }
}
```

Determine the **worst-case** time complexity.

12. Two algorithms have the following running times:

$$A(n) = 4n + 20$$

$$B(n) = 2n^2 + 5n + 10$$

a. Determine the asymptotic complexity of each.

b. Which algorithm is expected to be more efficient for large values of n?

c. Explain why.

13. A programmer has two sorting implementations:
    - **Algorithm A:** $2n^2$ instructions, running on a computer capable of 100 million instructions/second.
    - **Algorithm B:** $50n\log_2 n$ instructions, running on a computer capable of 1 million instructions/second.

    For $n = 100$, determine approximately which algorithm is faster.

14. Match each running time with its growth-rate class.

Running time	Class
7	?
$3\log n + 4$	?
$5n + 10$	?
$2n^2 + 3n$	?
$n^3 + 20n$	?
$2^n + n$	?